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To cite this article: Hayato Masuda, Takafumi Horie, Robert Hubacz, Naoto Ohmura & Makoto Shimoyamada (2017) Process development of starch hydrolysis using mixing characteristics of Taylor vortices, Bioscience, Biotechnology, and Biochemistry, 81:4, 755-761, DOI: [10.1080/09168451.2017.1282806](https://doi.org/10.1080/09168451.2017.1282806)

To link to this article: <https://doi.org/10.1080/09168451.2017.1282806>



Published online: 01 Feb 2017.



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Process development of starch hydrolysis using mixing characteristics of Taylor vortices

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Received November 1, 2016; accepted January 9, 2017
<http://dx.doi.org/10.1080/09168451.2017.1282806>

In food industries, enzymatic starch hydrolysis is an important process that consists of two steps: gelatinization and saccharification. One of the major difficulties in designing the starch hydrolysis process is the sharp change in its rheological properties. In this study, Taylor–Couette flow reactor was applied to continuous starch hydrolysis process. The concentration of reducing sugar produced via enzymatic hydrolysis was evaluated by varying operational variables: rotational speed of the inner cylinder, axial velocity (reaction time), amount of enzyme, and initial starch content in the slurry. When Taylor vortices were formed in the annular space, efficient hydrolysis occurred because Taylor vortices improved the mixing of gelatinized starch with enzyme. Furthermore, a modified inner cylinder was proposed, and its mixing performance was numerically investigated. The modified inner cylinder showed higher potential for enhanced mixing of gelatinized starch and the enzyme than the conventional cylinder.

Key words: starch hydrolysis process; food process design; Taylor–Couette flow reactor

Food engineering has become increasingly important because of the problems concerning environment and energy, e.g. decreasing availability of water and constraints on waste production. In order to establish high efficiency food processes, “conversion from batch to continuous operation” is significantly important.¹⁾ Continuous reactors are usually smaller than equivalent batch reactors. Furthermore, continuous operation leads to several benefits, such as improved control, better product quality, less hazardous operation, and reduced running costs. From the viewpoint of food engineering, one of the most difficult processes to establish a continuous process is starch hydrolysis process. Starch hydrolysis is an important process to hydrolyze starch

into glucose, which can be further converted to high fructose syrup for various applications in the food industry. Starch hydrolysis consists of two steps: gelatinization and saccharification. The fluid viscosity significantly increases through the gelatinization process, and the gelatinized starch is a non-Newtonian fluid having a shear-thinning property, i.e. its viscosity decreases with increasing shear-rate. However, the fluid viscosity drastically decreases through the enzymatic saccharification process. Due to the drastic changes in the viscosity during the gelatinization and saccharification processes, it is difficult to design a continuous process for starch hydrolysis. Furthermore, performance of the gelatinization process depends on heat transfer, while that of saccharification process depends on mixing of gelatinized starch with enzyme. This difference in key factors between gelatinization and saccharification makes it difficult to convert a batch process into a continuous one using a traditional stirred vessel. Furthermore, at high starch content in the slurry, conventional mixing operations with a stirred vessel are difficult because of high viscosity and strong shear-thinning property of the gelatinized starch. There have been many attempts to overcome these problems. For example, Kelder et al.²⁾ applied a curved tube to starch gelatinization and successfully showed that channeling phenomenon was suppressed by the Dean vortex. However, the starch content in the slurry was much lower (100 g/L) than that required in the industry (300–350 g/L). In the industry, a jet cooker is typically used for starch gelatinization, and a stirred vessel is used for saccharification.³⁾ Thus, at least two apparatuses are required for the starch hydrolysis process. For simultaneous gelatinization and saccharification at high starch content, many researchers, such as Reinikainen et al.,⁴⁾ Komolprasert and Ofoli,⁵⁾ Baks et al.,⁶⁾ and Emin and Schuchmann,⁷⁾ studied the application of extruders in the process. However, as extruders were developed for handling of dense gelatinized starch, they are not perfectly suitable for mixing gelatinized starch with

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enzyme in the saccharification process. Baks *et al.*⁶⁾ conducted saccharification using a stirred vessel after gelatinization. In order to improve the efficiency of continuous starch gelatinization and saccharification, Paolucci-Jeanjean *et al.*⁸⁾ proposed a membrane reactor. However, the starch content in the slurry was limited to 100 g/L.

The authors previously applied a Taylor–Couette flow reactor (TCFR)^{9,10)} to design a continuous process for starch gelatinization and saccharification using one apparatus. TCFR consists of two coaxial cylinders with the inner one rotating. Above a critical circumferential Reynolds number, Re_{cr} , pairs of counter-rotating toroidal vortices (Taylor vortices) appear, which are spaced regularly along the cylinder axis, as shown in Fig. 1. This toroidal vortex motion enhances not only heat and mass transfer, but also mixing. In addition, all fluid elements leaving the annulus have the same residence time on applying a relatively slow axial flow is applied.¹¹⁾ Therefore, the reactor enables converting a batch operation to a continuous one. The authors successfully showed that the formation of Taylor vortices in saccharification process was effective for enhanced mixing of gelatinized starch with enzyme at moderate starch content in the slurry (150 g/L).

In this study, the applicability of TCFR for more than 150 g/L starch content in the slurry, more than 150 g/L, was investigated. The reaction rate of starch gelatinization is faster, compared with the saccharification reaction.¹²⁾ Therefore, keeping the total volume of reactor constant, it is preferable to secure a larger volume of saccharification in TCFR than that of gelatinization. In this study, the shape of the inner cylinder was modified to be thick in the first section, corresponding to the gelatinization zone and thin in the second section, corresponding to the saccharification zone. To optimize the continuous process of starch hydrolysis using TCFR, mixing performance of the modified inner cylinder was evaluated and compared with that of the normal cylinder using numerical simulations.

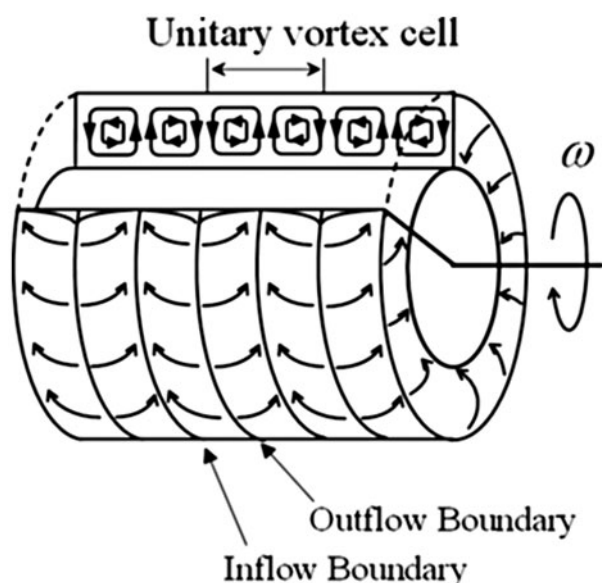


Fig. 1. Schematic picture of Taylor–Couette flow.

Materials and methods

Taylor–Couette flow reactor. The experimental apparatus (Fig. 2) consisted of concentric cylinders (length: 300 mm), rotating inner cylinder (outer radius: 12.5 mm), and fixed outer cylinder (inner radius: 17.5 mm) with a heating water-jacket. All parts of the apparatus were made of stainless steel. The starch suspended in water at 25 °C was introduced into the gap between the cylinders. The water jacket divided into two sections was employed for spatial temperature control of the processes; gelatinization requires high temperature, while the enzymatic saccharification process prefers moderate temperature. The temperature of the anterior half of the water jacket (gelatinization process: 0–150 mm) was 85 °C and that of the posterior half (saccharification process: 150–300 mm) was reduced to 45 °C. In general, starch slurry can be gelatinized completely at 85 °C for 5 min.¹²⁾ Therefore, at the enzyme injection point, starch slurry is assumed to be completely gelatinized. Thus, in this study, the effect of mixing by Taylor vortex flow on the viscosity can be limited to the saccharification section.¹¹⁾ Injection of an enzyme (α -amylase: derived from *Bacillus licheniformis*) in the middle point of the reactor resulted in continuous downstream starch hydrolysis. Initial starch content of the slurry, C_0 , was varied from 50 to 300 g/L. The axial velocity of starch slurry, u , was 0.00024 or 0.00048 m/s. The rotation speed of the inner cylinder, ω , was varied from 0 to 97 rad/s. The experimental conditions are summarized in Table 1. Furthermore, rheological properties (the shear viscosity) of the effluent were evaluated using a stress control rheometer (MCR-301, Anton Paar GmbH or HAAKE Viscotester 550, Thermo Fisher Scientific Inc.).

Measurement of sample. Samples of the product were taken at the outflow of the apparatus. In order to stop enzymatic saccharification, the samples were cooled immediately to about 0 °C, and HCl was added to acidify the samples to pH 2. The sample was centrifuged at 2000 g for more than 1800 s to separate the solid phase from the liquid phase. The concentration of the reducing sugar was determined using the dinitrosalicylic acid (DNS) method (2-hydroxy-3,5-

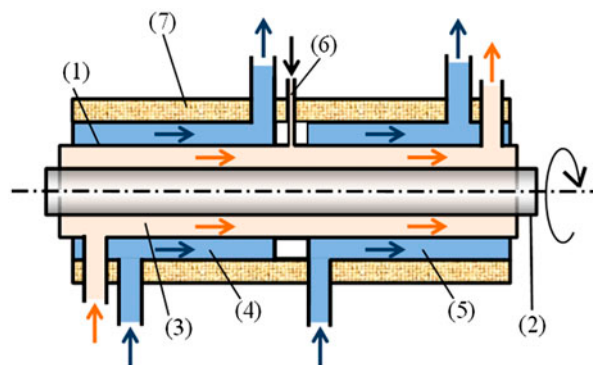


Fig. 2. Taylor–Couette flow reactor with water jackets: (1) Stationary outer cylinder (2) Rotating inner cylinder (3) Starch suspension (4) Hot water (5) Variable temperature water (6) Enzyme injection port (7) Insulator.

Table 1. Experimental condition.

Initial ungelatinized starch content C_0 [g/L]	50, 150, 300
Inlet axial velocity of starch slurry u_{ax} [cm/s]	0.024, 0.048
Rotation speed of the inner cylinder ω [rad/s]	0, 2, 5, 10, 15, 22, 47, 97
Inlet temperature T_{in} [°C]	25
Temperature of gelatinization process T_{ge} [°C]	85
Temperature of saccharification process T_{sa} [°C]	45
Enzyme volumetric flow rate v_e [mL/h]	0.1, 0.3, 0.6, 0.9, 1.8

Note: Starch from wheat (Wako Pure Chemical Industries, Ltd.) Termamyl 120L (α -amylase).

dinitrobenzoic acid, aqueous solution of 10 g/L 3,5-dinitrosalicylate, 16 g/L NaOH, and 300 g/L potassium sodium tartrate 4-hydrate). On addition of the DNS reagent, the sample developed an orange color, and concentration of the reducing sugar, C_{rs} , was measured using a UV-VIS spectrophotometer (MPS2400, SHIMADZU Co., Ltd.) at a wavelength of 540 nm.

Numerical simulation. Numerical simulation of fluid flow, including heat/mass transfer, is a powerful tool for understanding and optimizing the starch process. As described in the introduction section, modification of the shape of the inner cylinder would be beneficial for development of a new process. In order to investigate the applicability of the modified inner cylinder for starch hydrolysis, the mixing of gelatinized starch with the enzyme was simulated numerically. For comparison, numerical simulation using a normal cylinder was also conducted. Detailed information regarding the computational domain is shown in Fig. 3 for (a) normal inner cylinder and (b) modified inner cylinder.

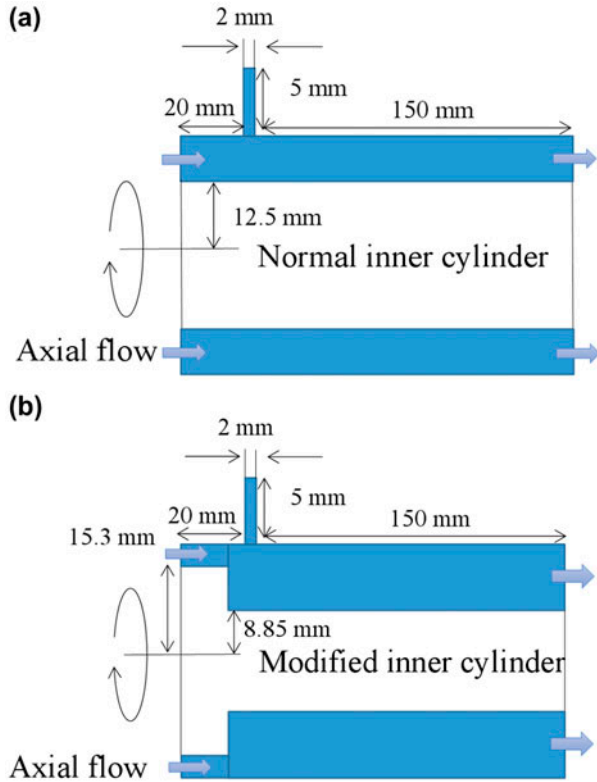


Fig. 3. Computational domain for (a) normal cylinder, and (b) modified inner cylinder.

The numerical simulation in this study was limited to the saccharification process (150 mm), including the run-up inlet zone (20 mm), which is a necessary part for the complete development of the fluid flow before contact of gelatinized starch with the enzyme. Similar to the experiment, the enzyme was introduced through the nozzle.

In this simulation, fluids were considered incompressible, and three-dimensional steady state was assumed. The governing equations, the conservation equation of mass, momentum, heat, and species should be solved to clarify transport phenomena in the reactor. The governing equations are as follows:

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{\nabla p}{\rho} + \frac{1}{\rho} \nabla \cdot (2\eta \mathbf{D}) + \mathbf{g} \quad (2)$$

$$\frac{\partial}{\partial t}(\rho C_p T) + \nabla \cdot (\rho C_p T \mathbf{u}) = \nabla \cdot (\lambda \nabla T) \quad (3)$$

$$\frac{\partial}{\partial t}(\rho X_j) + \nabla \cdot (\rho X_j \mathbf{u}) = \nabla \cdot (\rho D_j \nabla X_j) \quad (4)$$

where $\mathbf{u}$ [m/s] is the velocity, t [s] is the time, p [Pa] is the pressure, $\mathbf{g}$ [m/s²] is the gravitational acceleration, $\mathbf{D}$ [1/s] is the deformation rate tensor, C_p [J/kg K] is the specific heat capacity, T [K] is the temperature, λ [W/m K] is the thermal conductivity, X_j [-] is the mass fraction of the species j , and D_j [m²/s] is the diffusion coefficient of the species j . The viscous dissipation energy was neglected. As described by Pineda-Gómez et al.¹³⁾ and Hubacz et al.,⁹⁾ the values of C_p , λ , and D_j were chosen as 4,182 J/kg K, 0.6 W/m K and 10^{-13} m²/s, respectively. All simulations in this study were conducted using a commercial software (ANSYS 14.5 Fluent, Ansys Inc., United States). The second-order upwind scheme was used to discretize governing equations. The pressure-velocity coupling was performed using the SIMPLE scheme. It should be noted that viscosity change due to the reaction was neglected for simplicity. The complete simulation including the viscosity change due to the reaction will be conducted in the future.

The density and viscosity of the enzyme solution were assumed to be same as those of water, 1000 kg/m³ and 0.001 Pa s at 20 °C, respectively. Non-Newtonian properties of the effluent were modeled using the Carreau model¹⁴⁾:

$$\eta = (\eta_0 - \eta_\infty)[1 + (\beta \cdot \dot{\gamma})^2]^{(n-1)/2} + \eta_\infty \quad (5)$$

where η_0 is the zero shear-rate viscosity, η_∞ is the infinite shear-rate viscosity, $\dot{\gamma}$ is the shear-rate, β is the characteristic time, and n is the power index. The shear-rate is defined as $\dot{\gamma} = \sqrt{2\mathbf{D}:\mathbf{D}}$, which is the magnitude of the rate of deformation tensor. The sign after taking square root must be so chosen that $\dot{\gamma}$ is a positive quantity. As this Carreau model¹⁴ takes these features into account, the model is the most flexible among other rheological models, such as the power-law model. Furthermore, contrary to the power-law model that predicts an infinite viscosity at the limit of zero shear-rate, the Carreau model leads to a smooth transition to a constant viscosity at the limit of zero shear-rate, which is useful for numerical simulations. For example, rheological properties of effluents which were obtained by experiments for $C_0 = 300$ g/L are shown in Fig. 4, where Carreau model lines are also drawn. The model parameters used in the calculation are given in Table 2.

Results and discussion

Effect of Taylor vortex flow

In order to investigate the effect of Taylor vortex flow on starch saccharification, considerations based on Re in the circumferential direction are necessary. Re which expresses the flow condition, e.g. no vortex region or Taylor vortex flow region, is defined as follows:

$$Re = \frac{\rho R_i \omega d}{\eta} \quad (6)$$

where ρ [kg/m³] is the fluid density, R_i [m] is the inner cylinder radius, ω [rad/s] is the angular velocity of inner cylinder, d [m] is the gap width, and η [Pa s] is the fluid viscosity. Owing to the viscosity changes during saccharification reaction, the calculation of Re is difficult in this study. Figure 5 shows the relationship between the yield of reducing sugar, C_{rs}/C_0 , and rheological properties of the effluent at $C_0 = 50$ g/L and $T_{sa} = 45$ °C. Maintaining a constant residence time (i.e. reaction time), the yield was varied by controlling the volumetric flow rate of the enzyme. In Fig. 5, the viscosity dropped drastically from $C_{rs}/C_0 = 0$ to 0.11 and a further increase in C_{rs}/C_0 showed no significant decrease, suggesting that slight hydrolysis of gelatinized starch effectively affords the decrease in the

Table 2. Carreau model parameters at $C_0 = 300$ g/L.

	η_0 [Pa s]	η_∞ [Pa s]	n [-]	β [s]
$\omega = 10$ rad/s	0.5471	0	0.43	25
47 rad/s	0.746	0	0.28	1.6
97 rad/s	1.2	0	0.25	0.92

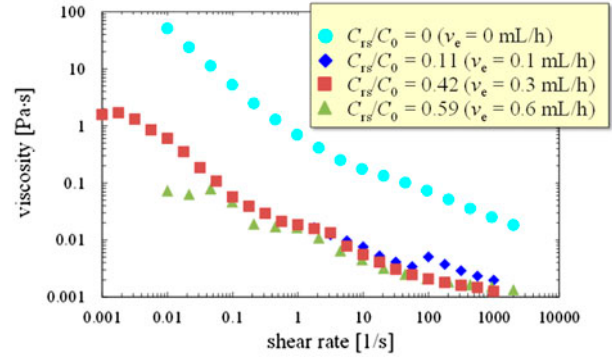


Fig. 5. Rheological properties of effluent at various v_e values ($C_0 = 50$ g/L, $\omega = 10$ rad/s).

fluid viscosity. Thus, the rheological property and density of the effluent can be used as the representative physical properties in the saccharification process. Furthermore, viscosities of all samples decreased with an increase in the shear-rate, and thus the effluent was considered to have a shear-thinning property. Owing to the shear-thinning property, the viscosity spatially changes with the spatial distribution of shear-rate in the annular space, as shown in Fig. 6. It should be noted that rheological parameters of the effluent at $C_0 = 300$ g/L and $\omega = 97$ rad/s were used in the numerical simulation. It is essential to estimate the effective shear-rate properly in order to calculate the effective Re (Re_{eff}) from the effective viscosity. Masuda et al.¹⁵ proposed an empirical correlation between the rotational speed of the inner cylinder (ω) and the effective shear-rate ($\dot{\gamma}_{eff}$) as follows,

$$\dot{\gamma}_{eff} = \left\{ 77.05n^{0.32} \left(\frac{R_i}{R_o} \right)^2 - 88.73n^{0.31} \left(\frac{R_i}{R_o} \right) + 26.85n^{0.21} \right\} \cdot \omega \quad (7)$$

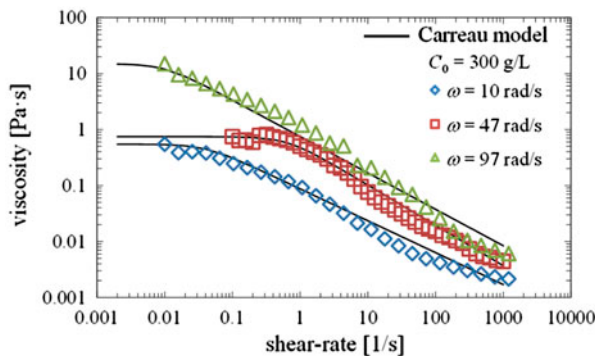


Fig. 4. Rheological properties of effluents, obtained from experiments with Carreau model lines.

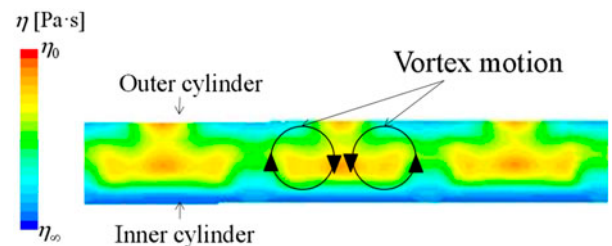


Fig. 6. Viscosity distribution in the annular space obtained by numerical simulation using rheological properties of effluent at $C_0 = 300$ g/L, $\omega = 97$ rad/s.

The value of n , which is a rheological parameter in Carreau model, could be obtained by fitting the rheological properties of the effluent to Carreau model. Using Equation (7), the representative shear-rate can be estimated from ω and n . From these considerations, the effective viscosity can be estimated to calculate Re_{eff} using Equation (6).

Figure 7 shows the dependency of C_{rs}/C_0 on Re_{eff} at $C_0 = 50\text{--}300\text{ g/L}$ and $T_{sa} = 45\text{ }^\circ\text{C}$. The enzyme-to-substrate ratio was unified as v_e : $C_0 = 0.006$: 1. The critical Re (Re_{cr}) for Taylor vortices is depicted as a dashed line at $Re_{cr} = 82$ ¹⁶⁾ in Fig. 7. As shown in Fig. 7, starch at high contents in the slurry was hydrolyzed effectively and continuously in the Taylor vortex flow region, where Re_{eff} was higher than 82. However, the C_{rs}/C_0 slightly decreased at higher Re_{eff} . This tendency can be explained by the wavy motion of Taylor vortices or enzyme inactivation by the high shear force. The wavy motion of Taylor vortices was reported in high Re_{eff} region.¹⁷⁾ The wavy motion negatively affected the mixing performance of Taylor vortices because enzyme may diffuse through the by-pass flow caused by the wavy motion of Taylor vortices.¹⁸⁾ This wavy motion occurs around $Re = 500$ in a Newtonian fluid system.¹⁹⁾ However, in a non-Newtonian fluid system, where the flow condition transits from laminar Taylor vortex flow to wavy vortex flow, Re_{eff} is not accurately determined. Therefore, establishment of flow transition based on Re_{eff} will be necessary. Another possibility is enzyme inactivation by the high shear force generated by high-speed rotation of the inner cylinder.²⁰⁾ For the optimization of TCFR for starch hydrolysis, the cause of the decrease in C_{rs}/C_0 at high Re_{eff} region should be investigated in the future. The result shown in Fig. 7 suggests that, in order to make the best use of the mixing characteristics of the Taylor vortex flow reactor, the reactor should be operated in the laminar Taylor vortex flow region. In addition, C_{rs}/C_0 obtained for $C_0 = 50\text{ g/L}$ using a stirred tank reactor with a Rushton turbine is shown in Fig. 7 for comparison. The volume of starch slurry and the enzyme, reaction temperature, and reaction time were set as well in the TCFR. The rotation speed of the impeller was set at 350 rpm, assumed to be sufficient

to achieve complete mixing. It should be noted that starch hydrolysis could not be conducted using the stirred tank reactor above $C_0 = 100\text{ g/L}$ because the uniform mixing of gelatinized starch of extremely high viscosity was very difficult. As shown in Fig. 7, C_{rs}/C_0 obtained using the TCFR under the laminar Taylor vortex flow region is equal to C_{rs}/C_0 obtained using the stirred tank reactor, i.e. batch reactor.

Figure 8 shows the effect of axial velocity on starch hydrolysis at $C_0 = 150\text{ g/L}$. Higher axial flow rate (i.e. higher u) means shorter residence time in TCFR. In case of $u = 0.048\text{ cm/s}$, within the experimental condition, C_{rs}/C_0 monotonically increased with increasing Re_{eff} without decrease in C_{rs}/C_0 at higher Re_{eff} . Many researchers reported that high u stabilizes the flow condition of Taylor vortices.^{21–23)} This means that, at higher u , the transition from laminar Taylor vortex flow to wavy vortex flow occurs at higher Re_{eff} than at lower u . Therefore, it is inferred that, at $u = 0.048\text{ cm/s}$, the wavy motion of Taylor vortices did not occur under the experimental conditions. However, future investigations based on fluid mechanics will be necessary to clarify the effect of axial velocity on C_{rs}/C_0 . Furthermore, Fig. 8 indicates that, in high Re_{eff} , starch slurry could be sufficiently hydrolyzed even in shorter residence time ($u = 0.048\text{ cm/s}$). For example, $C_{rs}/C_0 = 0.33$ could be obtained around $Re_{eff} = 1200$ even at $u = 0.048\text{ cm/s}$, while $C_{rs}/C_0 = 0.35$ could be obtained around $Re_{eff} = 100$ even at $u = 0.024\text{ cm/s}$. This result implies that higher rotation of the inner cylinder leads to reduced residence time.

Performance of modified inner cylinder

Figure 9 shows the contour of axial velocity at $\omega = 22\text{ rad/s}$ with modified inner cylinder obtained by numerical simulation. Rheological properties of the effluent at $C_0 = 300\text{ g/L}$ and $\omega = 22\text{ rad/s}$ were used for numerical simulation (Carreau model parameters: $\eta_0 = 0.6\text{ Pa s}$, $\eta_\infty = 0\text{ Pa s}$, $n = 0.4$, $\beta = 25\text{ s}$). Under the given conditions, Taylor vortices were not formed because of relatively low rotation speed of the inner cylinder. Nevertheless, a large vortex was observed at the contact zone of gelatinized starch and enzyme, enhancing the mixing of gelatinized starch with the enzyme. Figure 10 shows the enzyme distribution

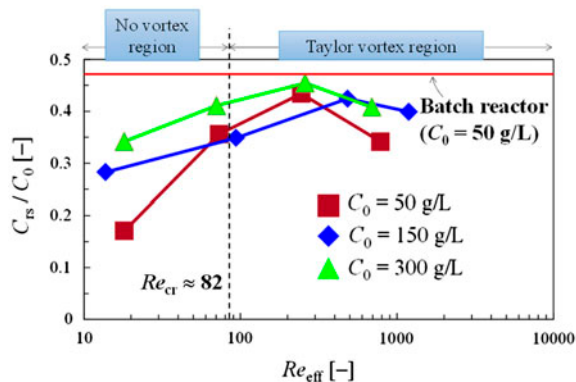


Fig. 7. Relation between concentration of reducing sugar and Re_{eff} at $C_0 = 50, 150, 300\text{ g/L}$, $u = 0.024\text{ cm/s}$, $T_{sa} = 45\text{ }^\circ\text{C}$.

Notes: The critical Re (Re_{cr}) for Taylor vortices is depicted as a dashed line at $Re_{cr} = 82$. Taylor vortices are formed when Re_{eff} is higher than 82.

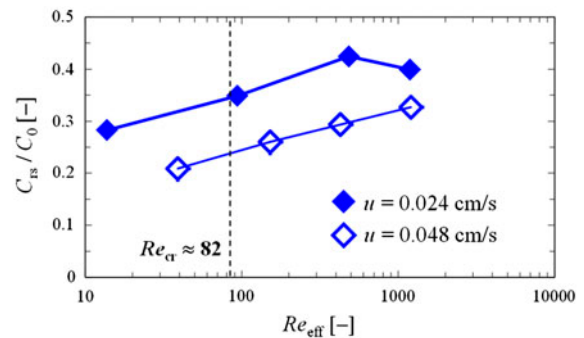


Fig. 8. Effect of axial velocity on concentration of reducing sugar at $C_0 = 150\text{ g/L}$.

Notes: The critical Re (Re_{cr}) for Taylor vortices is depicted as a dashed line at $Re_{cr} = 82$. Taylor vortices are formed when Re_{eff} is higher than 82.

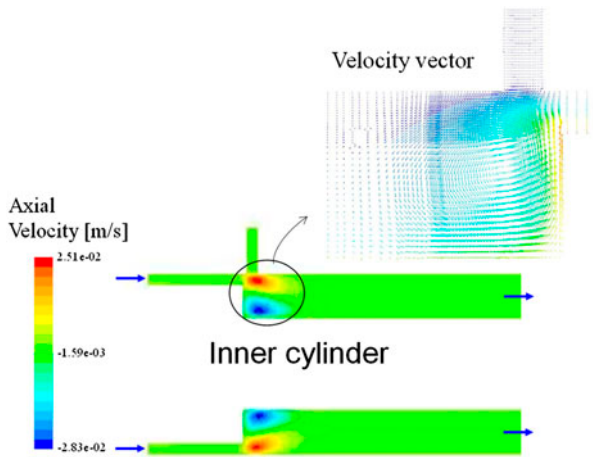


Fig. 9. Contour of axial velocity at $\omega = 22$ rad/s with modified inner cylinder.

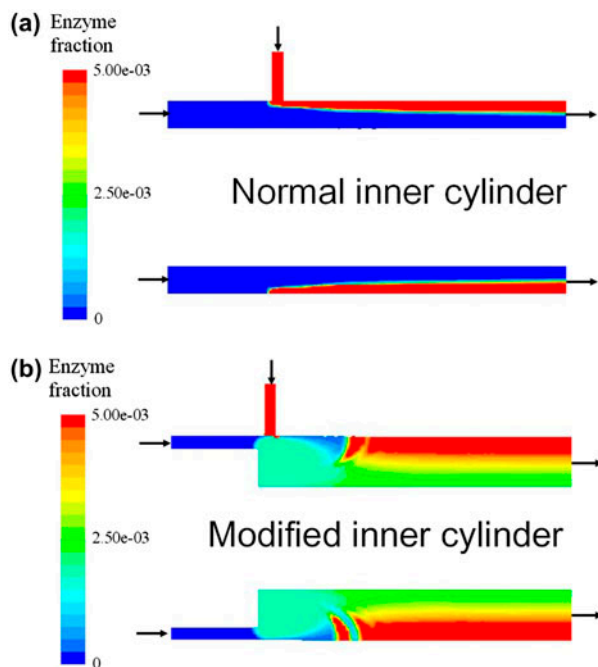


Fig. 10. Fraction of enzyme in the annular space at $\omega = 22$ rad/s with (a) normal cylinder and (b) modified inner cylinder.

simulated in the annular space at $\omega = 22$ rad/s with (a) normal cylinder and (b) modified inner cylinder, respectively. Figure 10(a) suggests that the mixing of gelatinized starch with the enzyme was insufficient due to limited diffusion in the radial direction. On the contrary, as clearly indicated in Fig. 10(b), the large vortex formed after the injection nozzle enhanced the mixing of gelatinized starch and enzyme. It can be speculated that the enzyme introduced from the nozzle is in direct contact with the gelatinized starch at the interface, triggering the saccharification reaction instantaneously. The enhanced mixing by the large vortex after the injection may play an important role in the effective saccharification reaction. Furthermore, the enhanced reaction might lead to a rapid decrease in the fluid viscosity. After the fluid viscosity decreases due to the saccharification reaction, Taylor vortices appear when Re_{eff} exceeds Re_{cr} . Therefore, it can be expected that Taylor vortices

in the modified inner cylinder can be formed at an early stage after the contact zone of gelatinized starch with enzyme, compared with the normal cylinder. This means that the number of Taylor vortices formed in the saccharification process for the modified inner cylinder would be more than that for the normal cylinder. As a result, mixing gelatinized starch with the enzyme might be further improved using the modified inner cylinder. Performance of the modified inner cylinder for starch hydrolysis will be experimentally investigated in the future.

Conclusion

In this study, Taylor–Couette flow reactor was applied to starch hydrolysis for the purpose of process development from the perspective of food engineering. The following conclusions were deduced:

- (1) In Taylor vortex flow region, starch slurry was hydrolyzed continuously and more efficiently.
- (2) At high Re_{eff} , starch slurry could be sufficiently hydrolyzed, even with shorter residence time.
- (3) The modified inner cylinder proposed in this study might enhance mixing gelatinized starch with the enzyme. The possible enhancement should be proved experimentally in future studies.

Author contributions

Masuda H, Horie T, Hubacz R, and Ohmura N conceived and designed the experiments and simulations. Masuda H and Hubacz R performed the experiments and simulations. Masuda H, Horie T, Hubacz R, and Ohmura N analyzed the data. Masuda H and Shimoyamada M wrote the manuscript.

Acknowledgments

The authors would like to thank Prof. Dr. Mitsuhiro Ohta (The University of Tokushima) for helpful discussion about numerical simulation, Dr. Yoshiyuki Komoda (Kobe University) for providing some laboratory and analytical equipment, and Mr. Norihisa Kumagai (Kobe University) for experimental support.

Disclosure statement

No potential conflict of interest was reported by the authors.

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